The Archive Is the Telescope: What Physics-Based PSF Subtraction Predicts for Hidden Worlds in the JWST Archive — and the Cold Census It Unlocks

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Author: Zubair Ali, Pharm-D student, SMBBMU Larkana, Pakistan · ORCID 0009-0001-5921-7509
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Abstract

Two results published in the first week of November 2026 change what existing telescope data can still deliver. First, Ferrer-Chavez et al. (2026) demonstrate Physics-based Learning for Imaging Planets (FLIP), a differentiable model-based point-spread-function subtraction that improves JWST/NIRCam coronagraphic sensitivity by a factor of 2–15 over the standard KLIP pipeline and recovers 51 Eridani b at 4.3 μm, where KLIP returned a non-detection. Second, Liu et al. (2026) show that "Capotauro", a candidate galaxy at redshift $z \simeq 32$, moved by 132 ± 20 milliarcseconds over 3.5 years and is instead a Y1 dwarf at $\sim$350 K and 730 ± 110 pc — one of the most distant Y dwarfs known. We combine these results with the JADES census of 41 T- and Y-dwarf candidates in 309.5 arcmin^2 and the two Bullet Cluster dropouts confirmed as Y dwarfs, and we derive three quantitative, falsifiable consequences. (1) FLIP-class reprocessing of the JWST coronagraphic archive: from the single published KLIP non-detection that FLIP turned into a 6.3σ detection, the one-sided 95% lower bound on the per-dataset recovery rate is 5%; reprocessing 20 such datasets therefore has a $\ge$64% probability of at least one new companion, and 59 datasets give $>$95%. (2) The Habitable Worlds Observatory stability requirement — 100–1000 times the stability of JWST — is reduced by the measured FLIP gain to a residual gap of 6.7–500 times, i.e. from 2–3 orders of magnitude to 0.8–2.7. (3) Capotauro's proper motion, 37.6 mas yr^{-1}, implies a transverse velocity of $\approx 130 \pm 20$ km s^{-1} at 730 pc, consistent with thick-disc/halo kinematics and with the JADES finding that most ultra-distant candidates are metal-poor subdwarfs. We state the observing programmes that would test each claim, including a 59-dataset falsification threshold for the archive yield and a proper-motion screen for the next ultra-high-redshift dropout candidates.

Keywords: exoplanets; direct imaging; JWST; PSF subtraction; brown dwarfs; Y dwarfs; high-redshift galaxies; Habitable Worlds Observatory

1. Introduction

Direct imaging of exoplanets is a fight against starlight. Even the brightest young giant planets are about 10^5 times fainter than their host stars in the near-infrared, so the stellar point spread function (PSF) must be modelled and subtracted to the required precision (Ferrer-Chavez et al. 2026). Until now the most successful approaches built the PSF model empirically, from reference observations of other stars or the same star at other times — and those approaches are fundamentally limited by telescope instability: pointing errors and thermal drifts change the aberrations between the reference and the science image, so the reference PSF never quite matches the science PSF. The limit this sets is called the stability limit, and it has bounded what direct imaging can do even from space.

On 4 November 2026, Ferrer-Chavez et al. published in *Nature* the first successful application of differentiable model-based PSF subtraction to real, on-sky coronagraphic data. Their framework, Physics-based Learning for Imaging Planets (FLIP), builds a physical model of the JWST Near-Infrared Camera (NIRCam), pre-calibrates it on a reference observation, and then solves — by

gradient descent, with $\sim 10^6$ free parameters — for the small changes in telescope aberrations that explain the science image. The result is a sensitivity gain of 2–15 times over the standard KLIP pipeline, largest at the smallest separations, and the recovery of a planet KLIP could not see at all. The data were already in the archive: the planet had been photographed two years earlier and missed.

In the same week, a second lesson arrived from the opposite end of astronomy. "Capotauro", a compact red source in the CEERS field proposed as a galaxy at redshift $z \simeq 32$ — light from $\sim$100 million years after the Big Bang — turned out to have moved. Liu et al. (2026) measured a shift of 132 ± 20 milliarcseconds over 3.5 years, ruling out any extragalactic source at $>6\sigma$, and identified the object as a Y-type brown dwarf of $\sim$350 K at 730 ± 110 pc. Two further F277W/F356W dropout candidates in the Bullet Cluster field, selected as possible $z \sim 20$–30 galaxies, were likewise shown by spectroscopy and proper motion to be Y dwarfs of 272–351 K and 445–525 K at $\sim$150–650 pc. And the JADES survey had already found 41 T- and Y-dwarf candidates in 309.5 arcmin^2 of deep extragalactic imaging, with Y dwarfs reaching 1–2 kpc (Hainline et al. 2026).

This paper does three things. First, it turns the single published FLIP recovery into a quantitative forecast for a systematic reprocessing of the JWST coronagraphic archive, with an explicit falsification threshold (§3.1). Second, it quantifies how much of the Habitable Worlds Observatory (HWO) stability problem FLIP-class processing removes (§3.2). Third, it derives the kinematics of Capotauro from its published proper motion and distance, and states what a proper-motion screen of the next ultra-high-redshift candidates should find (§3.3–§4).

2. Methods

2.1 Input values

All input numbers are taken from published sources, listed in Table 1. No new observational data are presented. The FLIP values are taken from Ferrer-Chavez et al. (2026, *Nature* 659, 58–62; DOI 10.1038/s41586-026-11213-z): sensitivity improvements of 2–15 times over KLIP; $\sim$15 times at 0.5 arcsec and $\sim$5 times at 1 arcsec for TYC 5899; $\sim$5 times at 0.5 arcsec and $\sim$2.5 times at 1 arcsec for 51 Eridani in F430M; $\sim$4 times at the planet's location in F430M; $1.6\times$ better signal-to-noise ratio (SNR) on HIP 65426 b; $2.1\times$ better SNR on 51 Eridani b in F410M; a SNR $= 6.3$ detection of 51 Eridani b in F430M where the original publication reported a non-detection; the first image of 51 Eridani b at 4.3 μm; $\sim 10^6$ free model parameters; $\sim$35 minutes per optimisation on a normal NVIDIA A30 GPU; and the HWO requirement of 10^{10} brightness contrast with picometre-level stability 100–1000 times that of JWST. The Capotauro values are taken from Liu et al. (2026, arXiv:2608.07461): a shift of 132 ± 20 mas over 3.5 years, proper motion $(\mu_\xi, \mu_\eta) = (+9.4, -35.6)$ mas yr^{-1} (total 37.6 mas yr^{-1}), spectral type Y1.0 $\pm$ 0.5, temperature $\approx$350 K, and distance 730 ± 110 pc. The Bullet Cluster dropout values are taken from the July 2026 preprint (arXiv:2604.23668): two candidates with Y-dwarf template temperatures of 272–351 K and 445–525 K, distances $\sim$150–650 pc, and proper motions of 49 ± 8 and 24 ± 3 mas yr^{-1}. The JADES values are taken from Hainline et al. (2026, arXiv:2510.00111): 41 brown-dwarf candidates (31 T, 10 Y) across 309.5 arcmin^2; T dwarfs out to 5–6 kpc, Y dwarfs out to 1–2 kpc; proper motions for nine candidates. The 51 Eridani system facts — planet discovered in December 2014 with the Gemini Planet Imager, announced August 2015; star $\sim$96 light-years away — are as reported in the discovery and press coverage of the FLIP result.

2.2 Calculations

All derived quantities in §3 were computed with a short Python script (`flip-archive-forecast-2026-11-07.py`, archived with this manuscript) and are reproducible from

the numbers in Table 1. The recovery-rate bound uses the exact one-sided Clopper–Pearson interval: with $k = 1$ success in $n = 1$ trial (the single KLIP non-detection dataset FLIP reprocessed that contained a planet), the 95% lower bound $p_{\rm low}$ satisfies $P(X \ge 1 \mid p_{\rm low}) = 0.05$, giving $p_{\rm low} = 0.05$. Detection probabilities for n reprocessed datasets assume independent trials at that rate: $P(\ge 1) = 1 - (1 - p)^n$. The falsification threshold uses the rule of three: zero successes in n trials bounds the rate at $p < 3/n$ at $\approx$95% confidence, so $n = 59$ is the smallest sample whose null result excludes $p \ge 0.05$ ($3/59 = 0.051$). The transverse velocity uses $v_t = 4.74\,\mu\,d$ km s^{-1} with μ in arcsec yr^{-1} and d in parsecs. The parallax at distance d is $1/d$ arcsec. Surface densities are candidate counts divided by the survey area. The HWO residual stability gap is the published 100–1000 requirement divided by the measured FLIP gain of 2–15.

3. Results

3.1 The archive yield: what reprocessing 20 datasets should find

FLIP was tested on three on-sky datasets. In the two known-planet systems it improved the SNR (HIP 65426 b by $1.6\times$, 51 Eridani b in F410M by $2.1\times$), and in F430M it turned a published KLIP non-detection into a 6.3σ detection of 51 Eridani b — the first image of the planet at 4.3 μm, in the carbon-dioxide absorption band that is inaccessible from the ground. On the third system, TYC 5899, which has no known companion, FLIP recovered injected fake planets that KLIP could not detect, with no false positive: the gain is real signal, not overfitting.

The single decisive event for forecasting is the F430M recovery: among KLIP non-detection datasets reprocessed so far, one out of one contained a missed planet. One event is a weak foundation, and this paper does not pretend otherwise — so we use the weakest reading the event supports. The one-sided 95% lower bound on the per-dataset recovery rate is $p \ge 0.05$. At that rate, reprocessing 20 KLIP non-detection NIRCam coronagraphic datasets gives $P(\ge 1\ {\rm recovery}) = 1 - 0.95^{20} = 0.64$, and 59 datasets give 0.95. In other words, even if the 51 Eridani recovery was near the bottom of what the evidence allows, a 20-dataset programme is better than even money to find a hidden companion, and a 59-dataset programme is a $>$95% bet. The compute cost is trivial: at $\sim$35 minutes per dataset on a single NVIDIA A30 GPU, 20 datasets cost $\sim$12 GPU-hours. The archive is the telescope, and the observing time is already paid for.

3.2 The HWO stability gap, reduced

Imaging an Earth analogue around a Sun-like star requires 10^{10} contrast, and the HWO concept therefore demands picometre-level stability 100–1000 times that of JWST — stability that, as Ferrer-Chavez et al. note, has not been achieved even in a laboratory. FLIP attacks exactly this requirement: by solving for the aberration changes instead of trusting a reference PSF, it circumvents the stability limit rather than engineering around it. Dividing the published requirement by the measured gain, the residual stability gap is $100/15 \approx 6.7$ in the best case (100$\times$ requirement, 15$\times$ gain) and $1000/2 = 500$ in the worst case — i.e. the gap shrinks from 2–3 orders of magnitude to 0.8–2.7 orders of magnitude. The gain is also largest exactly where it matters most: for TYC 5899 it is $\sim$15$\times$ at 0.5 arcsec versus $\sim$5$\times$ at 1 arcsec, and for 51 Eridani F430M $\sim$5$\times$ at 0.5 arcsec versus $\sim$2.5$\times$ at 1 arcsec — a factor of $\sim$2–3 improvement in the gain itself as the separation halves, towards the inner working angle where Earth-like planets orbit. This does not remove the stability problem: even the best case leaves a $\sim$7$\times$ gap, and FLIP's gains were measured on giant planets at 10^5 contrast, five orders of magnitude short of 10^{10}. But it changes the engineering target from "100–1000 times more stable than anything built" to "roughly one order of magnitude, with software doing the rest" — and it does so with a method already proven on real data, not a simulation.

3.3 Capotauro's kinematics, and the cold census

Capotauro's measured shift, 132 ± 20 mas over 3.5 years, is a proper motion of 37.7 mas yr^{-1} (consistent with the published total of 37.6 mas yr^{-1}). At the template distance of 730 ± 110 pc this implies a transverse velocity $v_t = 4.74 \times 0.0376 \times 730 \approx 130 \pm 20$ km s^{-1}, where the uncertainty is dominated by the $\pm$15% distance error. That velocity is far above the $\sim$20–40 km s^{-1} typical of thin-disc stars and is consistent with thick-disc or halo kinematics — the same population the JADES analysis infers for most of its ultra-distant candidates, which are best fit by sub-solar metallicity models. Capotauro's geometric parallax at 730 pc would be only 1.37 mas, well below the $\sim$20 mas astrometric uncertainty of the current measurement, which is why its distance remains photometric; a dedicated third epoch designed for parallax, as Liu et al. propose, is the way to turn the estimate into a measurement.

The census numbers put Capotauro in context. JADES found 41 candidates in 309.5 arcmin^2 — a surface density of 0.13 arcmin^{-2}, with Y dwarfs at 0.032 arcmin^{-2} — and 10 of them are Y dwarfs reaching 1–2 kpc. The Bullet Cluster field yielded two further Y dwarfs from two dropout candidates, with proper motions of 49 ± 8 and 24 ± 3 mas yr^{-1} detected on only a $\sim$1-year baseline. The coldest known brown dwarf, WISE J085510.83$-$071442.5 at $\sim$250 K, sits just 2 pc away; the instruments that found it could not have seen it at 730 pc. JWST deep fields are therefore, incidentally, the largest survey ever conducted of the coldest substellar population — and every ultra-high-redshift candidate list drawn from them carries a foreground of cold, moving dwarfs.

Quantity	Value	Source
FLIP sensitivity gain over KLIP	2–15×	Ferrer-Chavez et al. 2026, *Nature* 659, 58–62
FLIP gain, TYC 5899, 0.5″ / 1″	~15× / ~5×	Ferrer-Chavez et al. 2026
FLIP gain, 51 Eri F430M, 0.5″ / 1″	~5× / ~2.5×	Ferrer-Chavez et al. 2026
FLIP gain at 51 Eri b location, F430M	~4×	Ferrer-Chavez et al. 2026
51 Eri b, F430M (was KLIP non-detection)	SNR 6.3, first image at 4.3 μm	Ferrer-Chavez et al. 2026
FLIP SNR gain, HIP 65426 b / 51 Eri b F410M	1.6× / 2.1×	Ferrer-Chavez et al. 2026
FLIP optimisation cost	~10■ parameters, ~35 min, 1× NVIDIA A30	Ferrer-Chavez et al. 2026
HWO contrast / stability requirement	10¹■ / 100–1000× JWST	Ferrer-Chavez et al. 2026
Capotauro shift, 3.5-yr baseline	132 ± 20 mas	Liu et al. 2026, arXiv:2608.07461
Capotauro proper motion (total)	37.6 mas yr■¹	Liu et al. 2026
Capotauro type / temperature / distance	Y1.0 ± 0.5 / ~350 K / 730 ± 110 pc	Liu et al. 2026
Bullet Cluster dropouts: Teff / distance / μ	272–351 K, 445–525 K / ~150–650 pc / 49 ± 8, 24 ± 3 mas yr■¹	arXiv:2604.23668
JADES candidates / area	41 (31 T, 10 Y) / 309.5 arcmin²	Hainline et al. 2026, arXiv:2510.00111
JADES Y-dwarf reach	1–2 kpc	Hainline et al. 2026
Recovery rate, per KLIP non-detection (derived)	≥ 0.05 (95% lower bound)	This work, from the 1/1 event
P(≥1 recovery) in 20 / 59 datasets (derived)	0.64 / 0.95	This work
HWO residual stability gap (derived)	6.7–500× (0.8–2.7 orders)	This work
Capotauro v_t (derived)	≈ 130 ± 20 km s■¹	This work, from μ and d above
Capotauro parallax at 730 pc (derived)	1.37 mas	This work

Table 1. Input values from published sources and quantities derived in this work. Derived rows are labelled as such; all other values are quoted as published.

4. Discussion

4.1 The falsifiable claims

Claim 1 (archive yield). FLIP-class reprocessing of KLIP non-detection JWST/NIRCam coronagraphic datasets recovers missed companions at a per-dataset rate of at least 5%.

Prediction. Reprocessing the first 20 such datasets will yield at least one companion at SNR ≥ 5; the probability is $\ge$64% under the weakest reading of the published evidence.

Falsification. Zero recoveries in the first 59 reprocessed datasets would bound the rate at $p < 3/59 \approx 0.051$ at $\approx$95% confidence (rule of three), refuting the claim and showing the 51 Eridani recovery was exceptional rather than representative. A single recovery anywhere in the first 59 confirms the rate is at least of that order and justifies reprocessing the full archive.

Claim 2 (HWO stability). Physics-based PSF subtraction reduces the wavefront-stability requirement for Earth-analogue imaging by the measured sensitivity gain at the relevant separations, leaving a residual gap of one order of magnitude or less in the best case.

Prediction. Applying FLIP-class processing to simulated HWO-like coronagraphic data at 10^{10}-contrast-relevant separations will recover injected Earth-analogue signals at a stability budget relaxed by a factor of $\sim$2–15 relative to empirical reference subtraction.

Falsification. If the gain collapses below a factor of 2 at the separations and contrasts where HWO must operate — because model mismatch grows as the wavefront error shrinks, or because the $\sim 10^6$-parameter optimisation overfits at low SNR — then the stability requirement stands unreduced and the HWO engineering problem is unchanged.

Claim 3 (cold census). A substantial fraction of ultra-high-redshift ($z_{\rm phot} \gtrsim 20$) dropout candidates in JWST deep fields are foreground Y dwarfs, identifiable by proper motion alone on a $\sim$1–3.5-year baseline without spectroscopy.

Prediction. Among the next 10 dropout candidates at $z_{\rm phot} \gtrsim 20$ selected for second-epoch imaging, at least 2 will show proper motion >20 mas yr^{-1} — the scale already observed for Capotauro (37.6 mas yr^{-1}) and the Bullet Cluster dwarfs (49 ± 8 and 24 ± 3 mas yr^{-1}).

Falsification. If none of the next 10 candidates moves at >20 mas yr^{-1} on a $\ge$1-year baseline, the three moving dwarfs so far were a follow-up-selected fluke, and proper motion is not a useful screen for this population.

4.2 Why the two stories belong in one paper

FLIP and Capotauro are the same lesson from opposite directions. In both cases the data were already taken, and the discovery was in the processing: a planet photographed and missed because the PSF model trusted a reference image, and a "galaxy" believed for a year because nobody had measured whether it moved. The archive is full of both kinds of oversight — planets hidden under residual starlight, and nearby cold objects hiding among distant-galaxy candidates — and the tools that expose them are cheap: gradient descent on a physical instrument model, and a second epoch of imaging. Neither requires a new telescope. Both change what the next telescope must be: HWO's stability budget is smaller than assumed, and the next ultra-high-redshift claim must survive a proper-motion test before it survives peer review.

There is a direct operational link, too. The 4.3 μm detection of 51 Eridani b matters because that wavelength holds the carbon-dioxide absorption band, and CO$_2$ is a tracer of atmospheric

metallicity — the quantity that distinguishes formation pathways. Every companion FLIP recovers in the archive is therefore not only a detection but a potential atmospheric characterisation target at a wavelength ground-based telescopes cannot reach. The same logic extends to the cold census: Y dwarfs at 272–525 K are the temperature analogues of the coldest directly imaged planets, and a parallax-calibrated sample of them — the programme the JADES parallax result of 5 November 2026 begins — calibrates the atmospheric models that will be applied to planets FLIP finds.

4.3 What this paper does not claim

This paper presents no new data and no new measurement. The 5% recovery rate is a lower bound derived from a single event, not a measured rate; the true rate could be far higher (the only observed value is 1/1) or, if 51 Eridani was a favourable case, lower — which is exactly why the 59-dataset falsification threshold is stated. The HWO gap reduction divides a measured sensitivity gain by an engineering requirement; it assumes the gain transfers across five orders of magnitude of contrast, which is unproven and is itself the falsifiable part of Claim 2. The transverse velocity of Capotauro inherits the $\pm$15% photometric distance error and the assumption that the template match (Y1.0 $\pm$ 0.5, $\sim$350 K) is correct; the authors note the templates stop at Y1, so Capotauro could be colder, fainter and closer. The dropout-screen prediction (Claim 3) is based on a follow-up-selected sample of three objects, so its rate is an expectation for candidates selected for second-epoch imaging, not a population rate for all dropouts.

5. Conclusion

Two November 2026 results show that the most productive telescope of the next year may be the archive. FLIP improves JWST coronagraphic sensitivity by 2–15 times and has already recovered a planet from two-year-old data that standard processing missed; even the weakest reading of that single event implies a $\ge$64% chance of another recovery in the next 20 reprocessed datasets, at a cost of $\sim$12 GPU-hours, and zero recoveries in 59 datasets would refute the forecast outright. The same method cuts the Habitable Worlds Observatory stability gap from 2–3 orders of magnitude to 0.8–2.7. And the coldest objects in the sky keep turning up where galaxies were expected: Capotauro moved 132 ± 20 mas in 3.5 years, implying $\approx 130 \pm 20$ km s^{-1} of transverse motion at 730 pc, and the JADES census shows such objects are common in exactly the fields where the earliest galaxies are sought. Reprocess the archive, and re-image the candidates: both tests are cheap, both are decisive, and both can be run now.

Limitations

• The 5% recovery-rate bound rests on a single event (1 success in 1

trial); the Clopper–Pearson lower bound is exact but wide, and the true rate is essentially unconstrained above 0.05.

• The 20- and 59-dataset probabilities assume independent trials at a

constant rate; real archive datasets share instruments, filters and target selections, so the effective rate may vary with separation, filter and stellar type.

• The HWO gap reduction assumes the FLIP gain measured at 10^5 contrast

on giant planets transfers to 10^{10} contrast on Earth analogues; no evidence at that contrast exists yet.

• Capotauro's transverse velocity uses a photometric distance

(730 ± 110 pc) with no geometric parallax; a 1.37 mas parallax is below the current astrometric precision, so the distance — and hence the velocity — could shift with a dedicated parallax epoch.

• The dropout-screen prediction is calibrated on three follow-up-selected

objects; it is an expectation for candidates chosen for second-epoch imaging, not a population rate.

• This is an AI-assisted synthesis and calculation paper based on

published results; no new data were analysed and no author of the cited works was involved.

Data Availability

No new data were generated. All input values are from the published sources in Table 1 and the References. The calculation script (`flip-archive-forecast-2026-11-07.py`) is archived with this manuscript. arXiv identifiers and the Nature DOI cited were live and verified on 7 November 2026.

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References

• Ferrer-Chavez, R., Wang, J. J., Bouman, K. L., et al. (2026). Imaging exoplanets below the telescope stability limit. *Nature*, 659, 58–62. DOI: 10.1038/s41586-026-11213-z.

• Liu, F.-Y. F., McLeod, D. J., et al. (2026). A clear detection of proper motion confirms that the claimed $z \simeq 32$ galaxy candidate, "Capotauro", is a Y-type brown dwarf. *Monthly Notices of the Royal Astronomical Society*. arXiv:2608.07461.

• Hainline, K. N., et al. (2026). JADES: An Abundance of Ultra-Distant T- and Y-Dwarfs in Deep Extragalactic Data. arXiv:2510.00111.

• Brada■, M., Willott, C., Asada, Y., et al. (2026). Two Exciting High-redshift Galaxy Candidates Turn Out to Be Two Exciting Ultra-cool Brown Dwarfs. arXiv:2604.23668.

• Balmer, W. O., et al. (2025). JWST-TST high contrast: living on the wedge, or, NIRCam bar coronagraphy reveals CO$_2$ in the HR 8799 and 51 Eri exoplanets' atmospheres. *The Astronomical Journal*, 169, 209.

• Macintosh, B., et al. (2015). Discovery and spectroscopy of the young Jovian planet 51 Eri b with the Gemini Planet Imager. *Science*, 350, 64–67. DOI: 10.1126/science.aac5891.

• Luhman, K. L. (2014). Discovery of a $\sim$250 K brown dwarf at 2 pc from the Sun. *The Astrophysical Journal Letters*, 786, L18. DOI: 10.1088/2041-8205/786/2/L18.

• Boucher, S., et al. (2026). A Highly Reflective Atmosphere on the Lava World TOI-561b Revealed by JWST/NIRSpec Phase Curves. *The Astronomical Journal*. DOI: 10.3847/1538-3881/ae9763.

• Kirkpatrick, J. D., et al. (2021). The field substellar mass function based on the full-sky 20 pc census of 525 L, T, and Y dwarfs. *The Astrophysical Journal Supplement Series*, 253, 7. DOI: 10.3847/1538-4365/abd107.

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